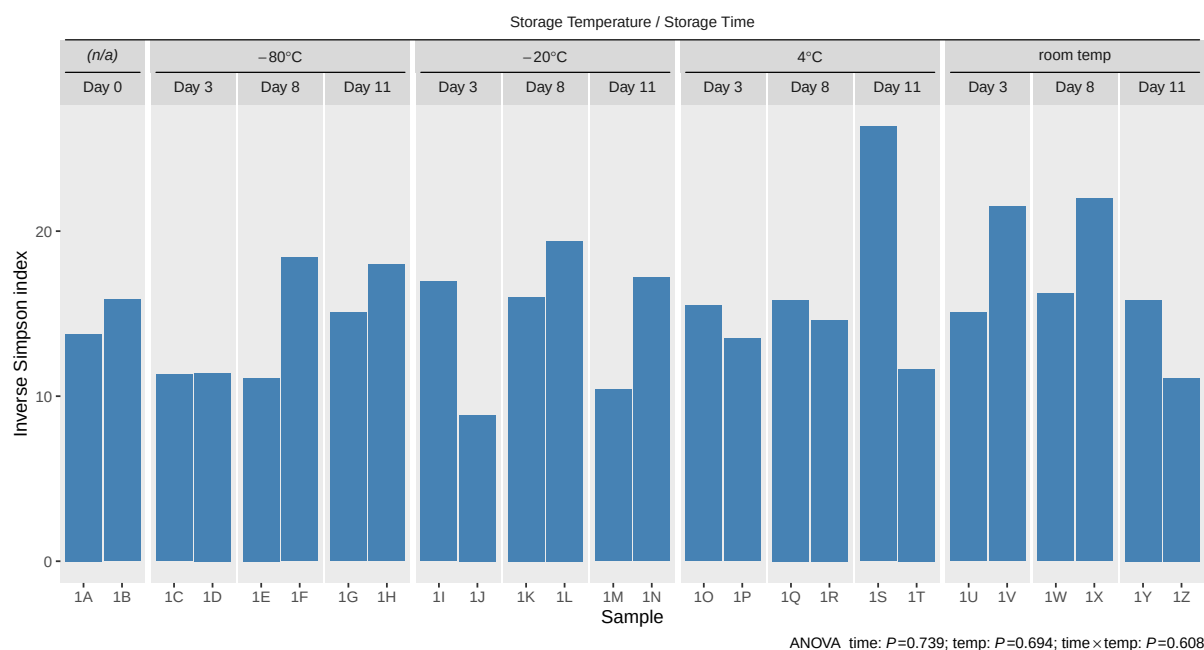
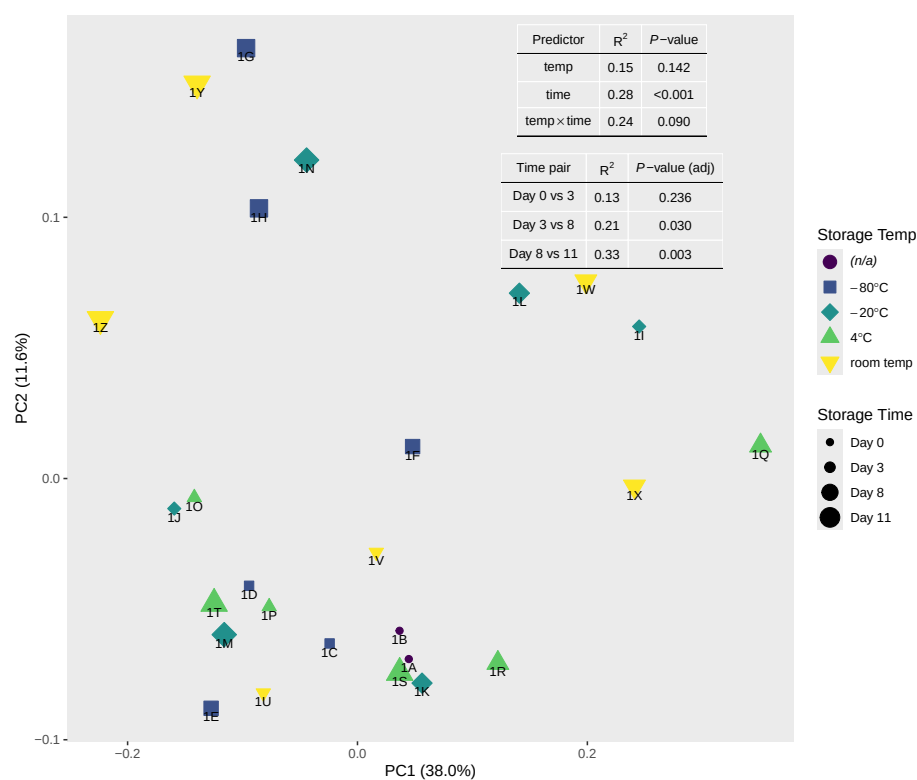


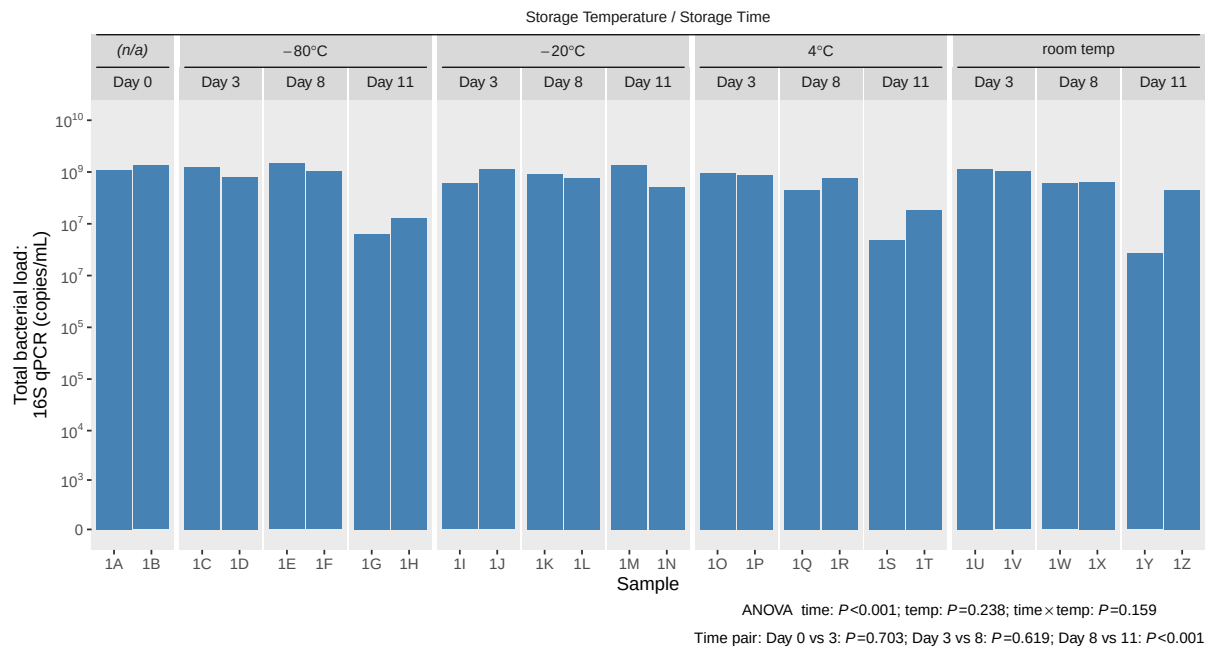
# Supplementary Materials



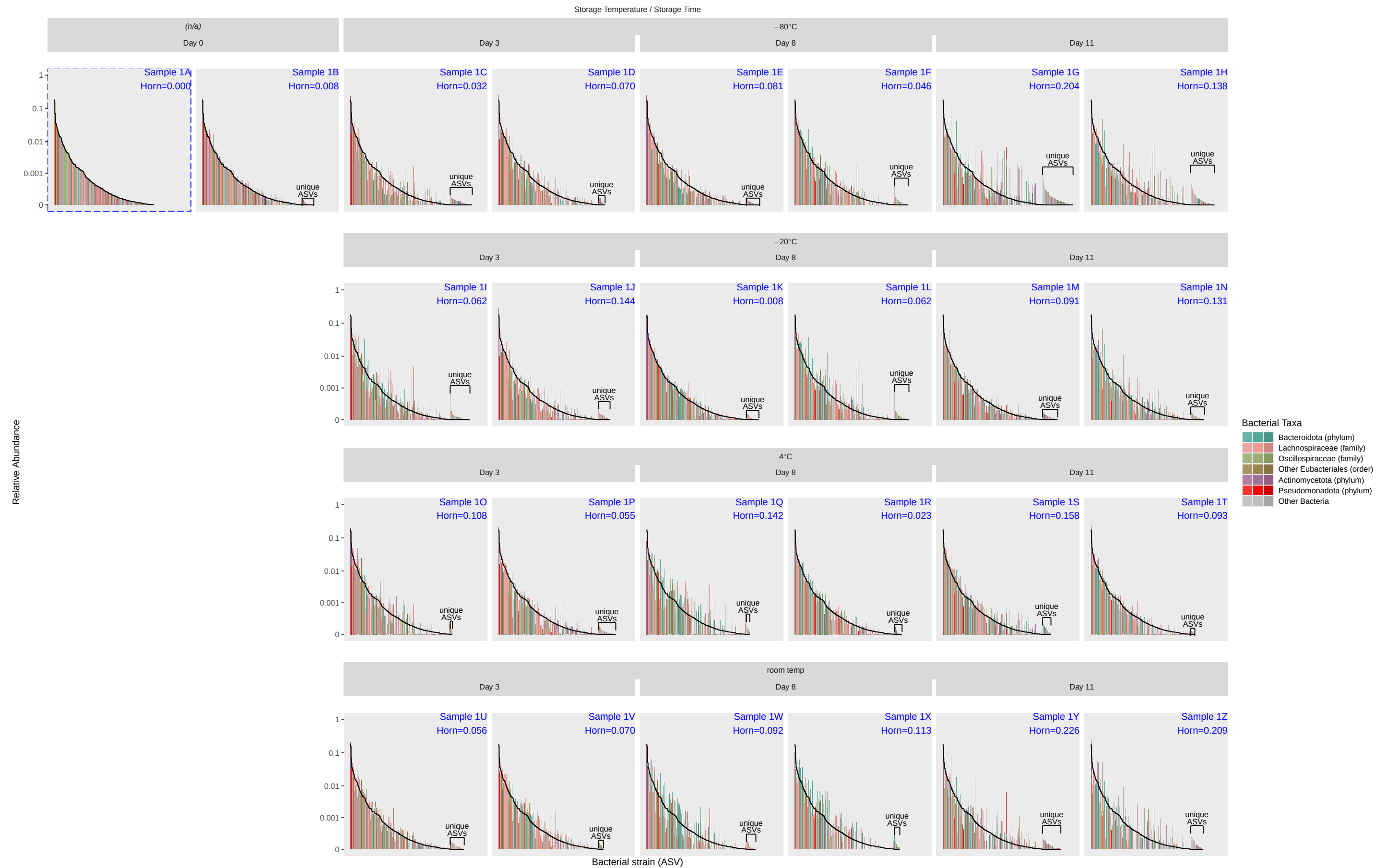
**Supplementary Figure S1:** Alpha diversity across storage conditions in the first experiment. Bar plots showing sample-level alpha diversity for the first experiment, as measured by the Inverse Simpson index. Samples are organized by storage temperature and storage duration, as indicated by the facet labels above each panel. ANOVA results for storage temperature, storage duration, and their interaction are shown below the plot.



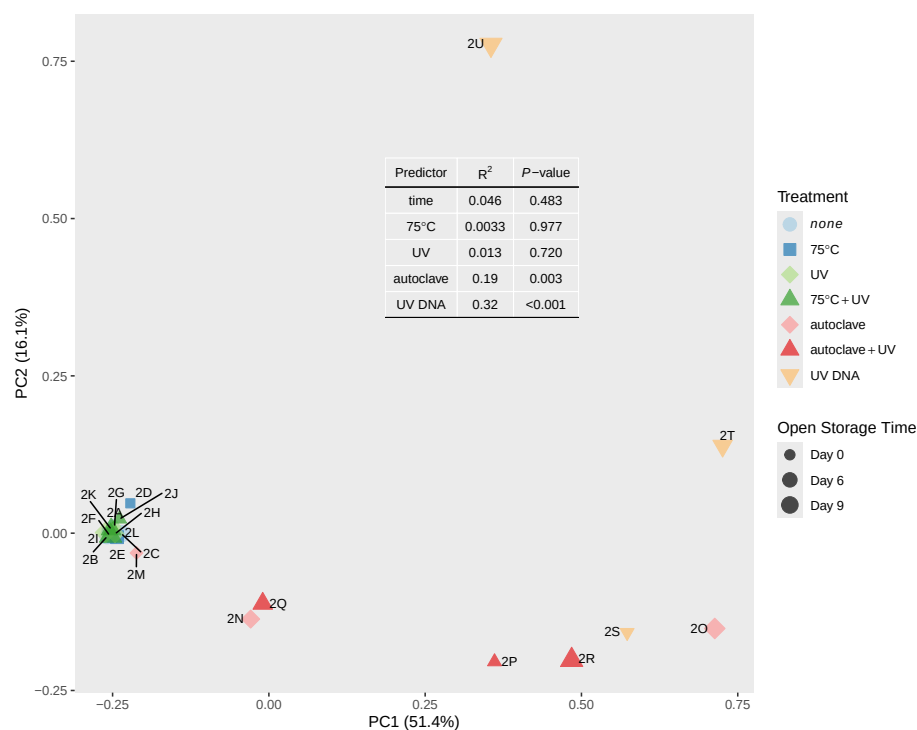
**Supplementary Figure S2:** Principal coordinate analysis of Bray–Curtis distances across storage conditions in the first experiment. Principal coordinate analysis (PCoA) plot based on Bray–Curtis distances between samples in the first experiment. Colors indicate storage duration, and shapes indicate storage temperature. Inset tables summarize PERMANOVA results for storage temperature, storage duration, and their interaction, along with pairwise comparisons between storage durations. Percent variation explained for each axis is indicated in parentheses.



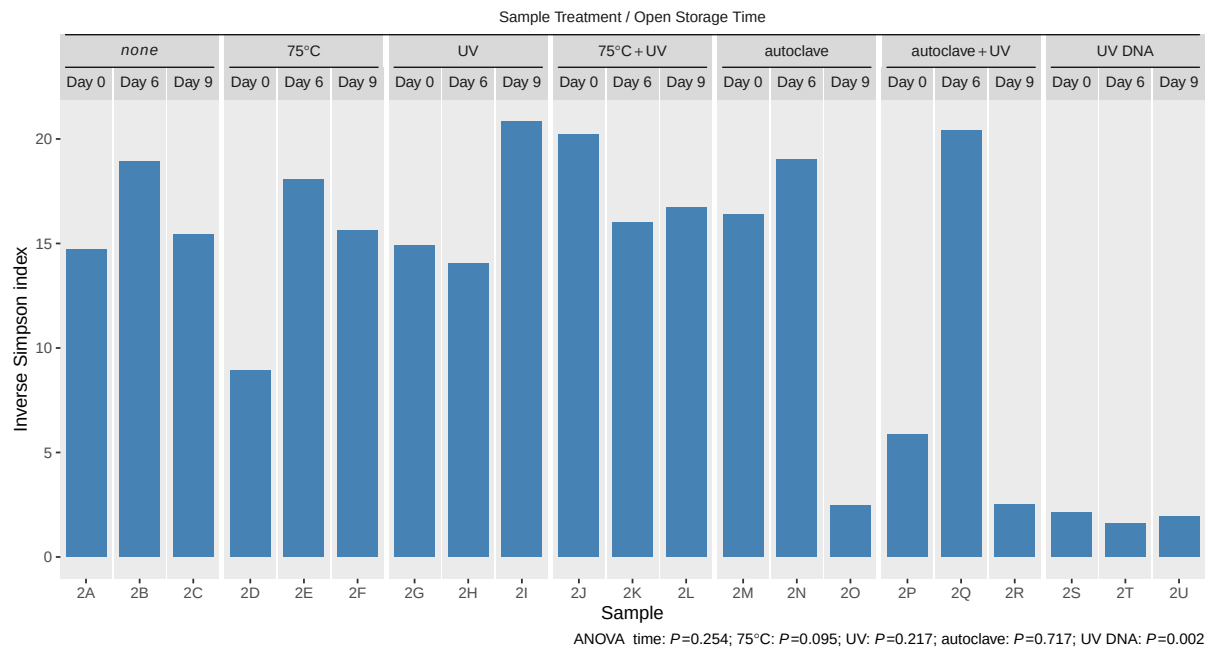
**Supplementary Figure S3:** Absolute bacterial abundance measured by 16S qPCR across storage conditions in the first experiment. Bar plot showing absolute bacterial abundance measured by 16S qPCR across all first-experiment samples. Samples are grouped by storage temperature and duration, as indicated by the facet labels above each panel. The y-axis is displayed on a logarithmic scale. Inset text summarizes ANOVA results for storage temperature, storage duration, and their interaction, along with pairwise comparisons between storage durations.



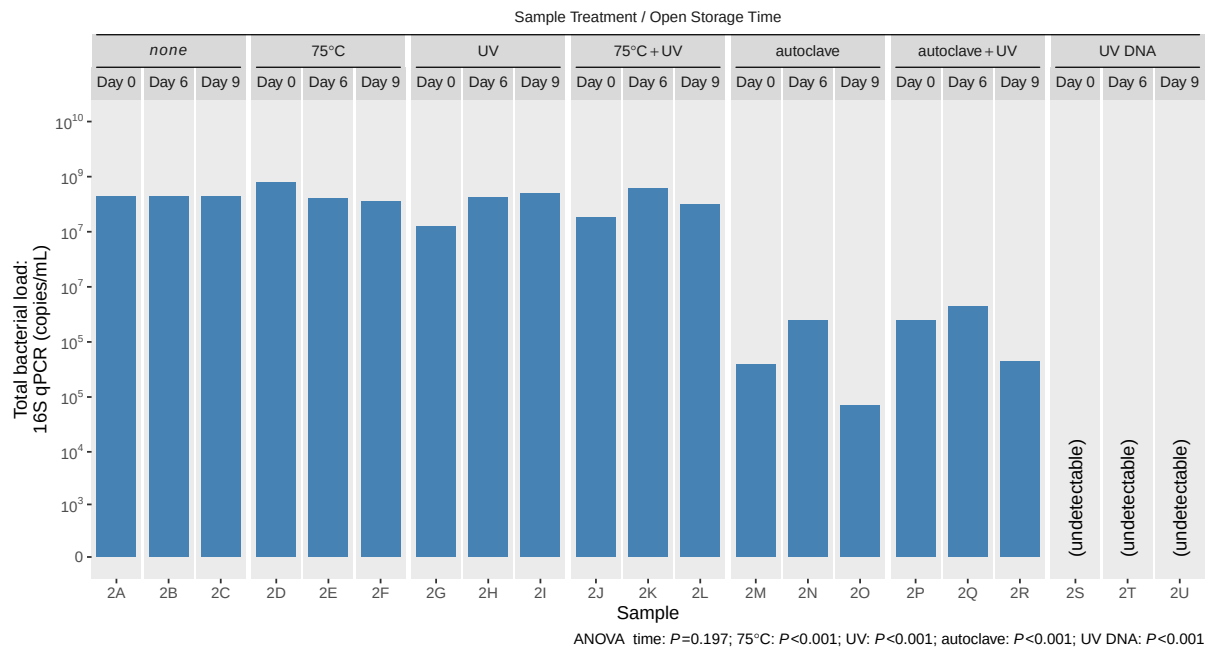
**Supplementary Figure S4:** Taxon-level relative abundance shifts relative to the baseline sample in the first experiment. Rank relative abundance plots showing taxon-level composition of all first experiment samples compared with the baseline sample, 1A (blue dashed box). Taxa are ordered by relative abundance in the baseline sample. Colored bars represent taxon abundances in each comparison sample, with colors indicating taxonomic classification. The black overlay line shows the baseline abundance profile for the same ordered taxa. Blue labels indicate the Morisita–Horn distance between each comparison sample and baseline.



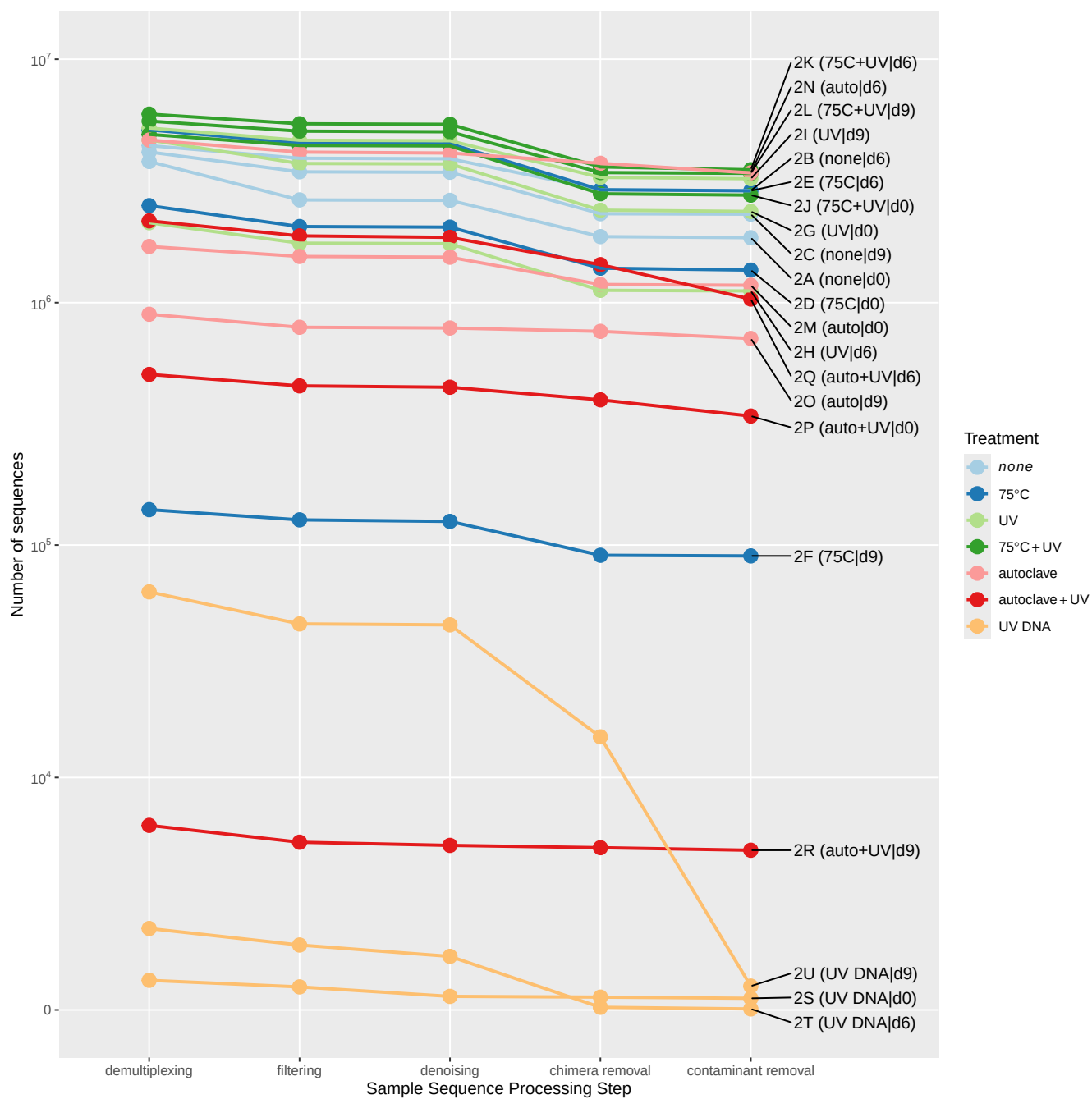
**Supplementary Figure S5:** Principal coordinate analysis of Bray–Curtis distances across storage durations and treatments in the second experiment. Principal coordinate analysis (PCoA) plot based on Bray–Curtis distances between samples in the second experiment. Colors indicate storage duration, and shapes indicate storage temperature. Inset tables summarize PERMANOVA results for 75 °C heat treatment, UV exposure before DNA extraction, UV exposure after DNA extraction, and pre-extraction autoclaving. Percent variation explained for each axis is indicated in parentheses.



**Supplementary Figure S6:** Alpha diversity across storage durations and treatments in the second experiment. Bar plots showing sample-level alpha diversity for the second experiment, as measured by the Inverse Simpson index. Samples are organized by treatment and storage duration, as indicated by the facet labels above each panel. ANOVA results evaluating the effects of storage duration, 75 °C heat treatment, UV exposure before DNA extraction, UV exposure after DNA extraction, and pre-extraction autoclaving are shown below the plot.



**Supplementary Figure S7:** Absolute bacterial abundance measured by 16S qPCR across storage durations and treatments in the second experiment. Bar plot showing absolute bacterial abundance measured by 16S qPCR across all second-experiment samples. Samples are grouped by treatment and storage duration, as indicated by the facet labels above each panel. ANOVA results evaluating the effects of storage duration, 75 °C heat treatment, UV exposure before DNA extraction, UV exposure after DNA extraction, and pre-extraction autoclaving are shown below the plot.



**Supplementary Figure S8:** Read retention across sequence-processing steps in the second experiment. Line plot showing the number of sequences retained for each second-experiment sample across sequential DADA2 processing steps, including demultiplexing, filtering, denoising, chimera removal, and contaminant removal. Lines represent individual samples and are colored by treatment condition. Sample labels are shown at the final processing step.



**Supplementary Table S1:** Linear discriminant analysis effect size (LEfSe) results comparing samples stored for 0–8 days with samples stored for 11 days. The Direction column indicates the group in which each taxon was enriched. Taxonomic assignments are shown with the corresponding rank, LDA score, and Kruskal–Wallis p-value. LDA, linear discriminant analysis; KW, Kruskal–Wallis

| Predictors of Storage Time |             |                                                                                                                           |  |  |  |  | LDA  | KW <i>P</i> -value | Log-Max |
|----------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------|--|--|--|--|------|--------------------|---------|
| Day 0-8                    | (4) Order   | Bacteria Bacillota Clostridia Eubacteriales                                                                               |  |  |  |  | 4.53 | 0.020              | 5.45    |
|                            | (5) Family  | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae                                                              |  |  |  |  | 4.63 | 0.006              | 5.37    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Faecalibacterium                                             |  |  |  |  | 4.58 | 0.009              | 5.29    |
|                            | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Faecalibacterium Faecalibacterium prausnitzii                |  |  |  |  | 4.58 | 0.009              | 5.29    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Ruminococcus                                                 |  |  |  |  | 3.45 | <0.001             | 4.06    |
|                            | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Ruminococcus Ruminococcus champanellensis                    |  |  |  |  | 3.45 | <0.001             | 4.06    |
|                            | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae — Clostridium leptum                                         |  |  |  |  | 3.06 | 0.030              | 3.81    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales — Gemmiger                                                                    |  |  |  |  | 3.71 | <0.001             | 4.42    |
|                            | (7) Species | Bacteria Bacillota Clostridia Eubacteriales — Gemmiger Gemmiger formicilis                                                |  |  |  |  | 3.71 | <0.001             | 4.42    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Agathobacter                                                 |  |  |  |  | 3.79 | 0.009              | 4.47    |
|                            | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Agathobacter Agathobacter rectalis                           |  |  |  |  | 3.79 | 0.009              | 4.47    |
|                            | (2) Phylum  | Bacteria Pseudomonadota                                                                                                   |  |  |  |  | 4.04 | 0.035              | 4.63    |
|                            | (3) Class   | Bacteria Pseudomonadota Betaproteobacteria                                                                                |  |  |  |  | 4.07 | 0.011              | 4.61    |
|                            | (4) Order   | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales                                                                |  |  |  |  | 4.07 | 0.011              | 4.61    |
|                            | (5) Family  | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae                                                 |  |  |  |  | 4.07 | 0.011              | 4.61    |
|                            | (6) Genus   | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae Parasutterella                                  |  |  |  |  | 4.07 | 0.011              | 4.61    |
|                            | (7) Species | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae Parasutterella Parasutterella excrementihominis |  |  |  |  | 4.07 | 0.011              | 4.61    |
| Day 11                     | (7) Species | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae Bifidobacterium Bifidobacterium longum         |  |  |  |  | 4.12 | 0.023              | 4.96    |
|                            | (5) Family  | Bacteria Bacillota Clostridia Eubacteriales Clostridiaceae                                                                |  |  |  |  | 4.15 | 0.005              | 4.60    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Clostridiaceae Clostridium                                                    |  |  |  |  | 4.15 | 0.006              | 4.60    |
|                            | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Clostridiaceae Clostridium Clostridium innocuum                               |  |  |  |  | 4.15 | 0.005              | 4.59    |
|                            | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Tyzzerella                                                   |  |  |  |  | 4.17 | 0.035              | 4.73    |
|                            | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Tyzzerella Clostridium nexile                                |  |  |  |  | 4.17 | 0.035              | 4.73    |
|                            | (3) Class   | Bacteria Bacillota Erysipelotrichia                                                                                       |  |  |  |  | 3.90 | 0.046              | 4.64    |
|                            | (4) Order   | Bacteria Bacillota Erysipelotrichia Erysipelotrichales                                                                    |  |  |  |  | 3.90 | 0.046              | 4.64    |
|                            | (5) Family  | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Coprobacillaceae                                                   |  |  |  |  | 3.89 | 0.035              | 4.58    |
|                            | (6) Genus   | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Coprobacillaceae Coprobacillus                                     |  |  |  |  | 3.33 | 0.023              | 3.85    |
|                            | (7) Species | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Coprobacillaceae Coprobacillus Coprobacillus cateniformis          |  |  |  |  | 3.33 | 0.023              | 3.85    |
|                            | (3) Class   | Bacteria Bacillota Tissierellia                                                                                           |  |  |  |  | 3.32 | 0.035              | 3.73    |
|                            | (4) Order   | Bacteria Bacillota Tissierellia Tissierellales                                                                            |  |  |  |  | 3.32 | 0.035              | 3.73    |
|                            | (5) Family  | Bacteria Bacillota Tissierellia Tissierellales Peptoniphilaceae                                                           |  |  |  |  | 3.32 | 0.035              | 3.73    |

**Supplementary Table S2:** Linear discriminant analysis effect size (LEfSe) results are shown for two comparisons: autoclaved samples versus untreated samples, and UV-DNA irradiated samples versus untreated samples. The Direction column indicates the group in which each taxon was enriched. Taxonomic assignments are shown with the corresponding rank, LDA score, and Kruskal–Wallis p-value. LDA, linear discriminant analysis; KW, Kruskal–Wallis.

| Predictors of autoclave vs. no treatment |             |                                                                                                                           |      |    |         |         |
|------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------|------|----|---------|---------|
| Direction                                | Rank        | Taxonomy                                                                                                                  | LDA  | KW | P-value | Log-Max |
| autoclave                                | (3) Class   | Bacteria Bacteroidota Flavobacteriia                                                                                      | 3.22 |    | 0.037   | 3.52    |
|                                          | (4) Order   | Bacteria Bacteroidota Flavobacteriia Flavobacteriales                                                                     | 3.22 |    | 0.037   | 3.52    |
|                                          | (5) Family  | Bacteria Bacteroidota Flavobacteriia Flavobacteriales Weeksellaceae                                                       | 3.22 |    | 0.037   | 3.52    |
|                                          | (6) Genus   | Bacteria Bacteroidota Flavobacteriia Flavobacteriales Weeksellaceae Chryseobacterium                                      | 3.12 |    | 0.037   | 3.42    |
|                                          | (7) Species | Bacteria Bacteroidota Flavobacteriia Flavobacteriales Weeksellaceae Chryseobacterium Chryseobacterium rhizosphaerae       | 3.12 |    | 0.037   | 3.42    |
|                                          | (2) Phylum  | Bacteria Pseudomonadota                                                                                                   | 5.28 |    | 0.050   | 5.63    |
|                                          | (5) Family  | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Alcaligenaceae                                                 | 5.14 |    | 0.050   | 5.45    |
|                                          | (6) Genus   | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Alcaligenaceae Achromobacter                                   | 5.14 |    | 0.050   | 5.45    |
|                                          | (7) Species | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Alcaligenaceae Achromobacter Achromobacter denitrificans       | 5.14 |    | 0.050   | 5.45    |
|                                          | (3) Class   | Bacteria Pseudomonadota Gammaproteobacteria                                                                               | 4.80 |    | 0.050   | 5.10    |
|                                          | (4) Order   | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales                                                              | 4.46 |    | 0.050   | 4.77    |
|                                          | (5) Family  | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae                                                 | 4.46 |    | 0.037   | 4.76    |
|                                          | (6) Genus   | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae Ewingella                                       | 4.36 |    | 0.037   | 4.66    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae Ewingella Ewingella americana                   | 4.36 |    | 0.037   | 4.66    |
|                                          | (6) Genus   | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae Serratia                                        | 3.79 |    | 0.037   | 4.09    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae Serratia Serratia liquefaciens                  | 3.64 |    | 0.037   | 3.94    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Enterobacterales Yersiniaceae Serratia Serratia marcescens                    | 3.27 |    | 0.037   | 3.57    |
|                                          | (4) Order   | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales                                                               | 4.52 |    | 0.037   | 4.83    |
|                                          | (5) Family  | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae                                              | 4.52 |    | 0.037   | 4.83    |
|                                          | (6) Genus   | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas                                  | 4.52 |    | 0.037   | 4.83    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas Pseudomonas fluorescens          | 4.19 |    | 0.037   | 4.49    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas Pseudomonas lini                 | 3.04 |    | 0.037   | 3.34    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas Pseudomonas palleroniana         | 3.56 |    | 0.037   | 3.86    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas Pseudomonas protegens            | 4.06 |    | 0.037   | 4.36    |
|                                          | (7) Species | Bacteria Pseudomonadota Gammaproteobacteria Pseudomonadales Pseudomonadaceae Pseudomonas Pseudomonas veronii              | 3.27 |    | 0.037   | 3.57    |
| none                                     | (2) Phylum  | Bacteria Actinomycetota                                                                                                   | 4.63 |    | 0.050   | 5.07    |
|                                          | (3) Class   | Bacteria Actinomycetota Actinomycetes                                                                                     | 4.60 |    | 0.050   | 5.03    |
|                                          | (4) Order   | Bacteria Actinomycetota Actinomycetes Bifidobacteriales                                                                   | 4.60 |    | 0.050   | 5.03    |
|                                          | (5) Family  | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae                                                | 4.60 |    | 0.050   | 5.03    |
|                                          | (6) Genus   | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae Bifidobacterium                                | 4.60 |    | 0.050   | 5.03    |
|                                          | (7) Species | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae Bifidobacterium Bifidobacterium longum         | 4.60 |    | 0.050   | 5.03    |
|                                          |             |                                                                                                                           |      |    |         |         |
| Predictors of UV DNA vs. no treatment    |             |                                                                                                                           |      |    |         |         |
| Direction                                | Rank        | Taxonomy                                                                                                                  | LDA  | KW | P-value | Log-Max |
| UV DNA                                   | (1) Domain  | Bacteria                                                                                                                  | 4.15 |    | 0.037   | 6.00    |
|                                          | (2) Phylum  | Bacteria Pseudomonadota                                                                                                   | 5.65 |    | 0.046   | 5.97    |
| none                                     | (2) Phylum  | Bacteria Actinomycetota                                                                                                   | 4.77 |    | 0.037   | 5.07    |
|                                          | (3) Class   | Bacteria Actinomycetota Actinomycetes                                                                                     | 4.73 |    | 0.037   | 5.03    |
|                                          | (4) Order   | Bacteria Actinomycetota Actinomycetes Bifidobacteriales                                                                   | 4.73 |    | 0.037   | 5.03    |
|                                          | (5) Family  | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae                                                | 4.73 |    | 0.037   | 5.03    |
|                                          | (6) Genus   | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae Bifidobacterium                                | 4.73 |    | 0.037   | 5.03    |
|                                          | (7) Species | Bacteria Actinomycetota Actinomycetes Bifidobacteriales Bifidobacteriaceae Bifidobacterium Bifidobacterium longum         | 4.73 |    | 0.037   | 5.03    |
|                                          | (3) Class   | Bacteria Actinomycetota Coriobacteriia                                                                                    | 3.71 |    | 0.037   | 4.01    |
|                                          | (4) Order   | Bacteria Actinomycetota Coriobacteriia Coriobacteriales                                                                   | 3.67 |    | 0.037   | 3.98    |
|                                          | (5) Family  | Bacteria Actinomycetota Coriobacteriia Coriobacteriales Coriobacteriaceae                                                 | 3.67 |    | 0.037   | 3.98    |
|                                          | (6) Genus   | Bacteria Actinomycetota Coriobacteriia Coriobacteriales Coriobacteriaceae Collinsella                                     | 3.67 |    | 0.037   | 3.98    |
|                                          | (7) Species | Bacteria Actinomycetota Coriobacteriia Coriobacteriales Coriobacteriaceae Collinsella Collinsella aerofaciens             | 3.67 |    | 0.037   | 3.98    |
|                                          | (2) Phylum  | Bacteria Bacillota                                                                                                        | 5.54 |    | 0.046   | 5.87    |
|                                          | (3) Class   | Bacteria Bacillota Clostridia                                                                                             | 5.53 |    | 0.046   | 5.86    |
|                                          | (4) Order   | Bacteria Bacillota Clostridia Eubacteriales                                                                               | 5.16 |    | 0.046   | 5.52    |
|                                          | (5) Family  | Bacteria Bacillota Clostridia Eubacteriales Butyricicoccaceae                                                             | 3.79 |    | 0.037   | 4.09    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Butyricicoccaceae Agathobaculum                                               | 3.79 |    | 0.037   | 4.09    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Butyricicoccaceae Agathobaculum Agathobaculum butyriciproducens               | 3.79 |    | 0.037   | 4.09    |
|                                          | (5) Family  | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae                                                              | 5.10 |    | 0.046   | 5.47    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Drancourtella                                                | 3.05 |    | 0.037   | 3.35    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Drancourtella Drancourtella massiliensis                     | 3.05 |    | 0.037   | 3.35    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Faecalibacterium                                             | 5.08 |    | 0.046   | 5.46    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Faecalibacterium Faecalibacterium prausnitzii                | 5.08 |    | 0.046   | 5.46    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Flavonifractor                                               | 3.22 |    | 0.037   | 3.52    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Flavonifractor Flavonifractor plautii                        | 3.22 |    | 0.037   | 3.52    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Oscillibacter                                                | 3.07 |    | 0.037   | 3.38    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales Oscillospiraceae Oscillibacter Oscillibacter ruminantium                      | 3.07 |    | 0.037   | 3.38    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Eubacteriales — Gemmiger                                                                    | 4.02 |    | 0.037   | 4.32    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Eubacteriales — Gemmiger Gemmiger formicilis                                                | 4.02 |    | 0.037   | 4.32    |
|                                          | (4) Order   | Bacteria Bacillota Clostridia Lachnospirales                                                                              | 5.29 |    | 0.046   | 5.60    |
|                                          | (5) Family  | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae                                                              | 5.29 |    | 0.046   | 5.60    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Agathobacter                                                 | 4.65 |    | 0.037   | 4.95    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Agathobacter Agathobacter rectalis                           | 4.65 |    | 0.037   | 4.95    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Anaerobutyricum                                              | 3.85 |    | 0.037   | 4.15    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Anaerobutyricum Anaerobutyricum hallii                       | 3.85 |    | 0.037   | 4.15    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Anaerostipes                                                 | 4.52 |    | 0.046   | 4.88    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Anaerostipes Anaerostipes hadrus                             | 4.52 |    | 0.046   | 4.88    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia                                                      | 4.88 |    | 0.037   | 5.18    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia Blautia faecis                                       | 3.57 |    | 0.037   | 3.87    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia Blautia luti                                         | 4.68 |    | 0.037   | 4.99    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia Blautia obeum                                        | 3.86 |    | 0.037   | 4.16    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia Blautia stercoris                                    | 3.63 |    | 0.037   | 3.93    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Blautia Blautia wexlerae                                     | 4.02 |    | 0.037   | 4.32    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Dorea                                                        | 3.92 |    | 0.037   | 4.22    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Dorea Dorea longicatena                                      | 3.92 |    | 0.037   | 4.22    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Fusicatenibacter                                             | 4.11 |    | 0.037   | 4.41    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Fusicatenibacter Fusicatenibacter saccharivorans             | 4.11 |    | 0.037   | 4.41    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Herbinix                                                     | 3.25 |    | 0.037   | 3.55    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Herbinix Herbinix luporum                                    | 3.25 |    | 0.037   | 3.55    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Kineothrix                                                   | 3.10 |    | 0.037   | 3.40    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Kineothrix Kineothrix alysoides                              | 3.10 |    | 0.037   | 3.40    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Lachnospira                                                  | 3.05 |    | 0.037   | 3.35    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Lacrimispora                                                 | 3.08 |    | 0.037   | 3.38    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Mediterraneibacter                                           | 3.69 |    | 0.037   | 3.99    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Mediterraneibacter Mediterraneibacter faecis                 | 3.27 |    | 0.037   | 3.57    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Mediterraneibacter Mediterraneibacter gnavus                 | 3.41 |    | 0.037   | 3.71    |
|                                          | (6) Genus   | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Roseburia                                                    | 3.48 |    | 0.037   | 3.78    |
|                                          | (7) Species | Bacteria Bacillota Clostridia Lachnospirales Lachnospiraceae Roseburia Roseburia hominis                                  | 3.41 |    | 0.037   | 3.71    |
|                                          | (3) Class   | Bacteria Bacillota Erysipelotrichia                                                                                       | 3.33 |    | 0.037   | 3.63    |
|                                          | (4) Order   | Bacteria Bacillota Erysipelotrichia Erysipelotrichales                                                                    | 3.33 |    | 0.037   | 3.63    |
|                                          | (5) Family  | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Erysipelotrichaceae                                                | 3.32 |    | 0.037   | 3.62    |
|                                          | (6) Genus   | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Erysipelotrichaceae Holdemania                                     | 3.28 |    | 0.037   | 3.58    |
|                                          | (7) Species | Bacteria Bacillota Erysipelotrichia Erysipelotrichales Erysipelotrichaceae Holdemania Holdemania filiformis               | 3.28 |    | 0.037   | 3.58    |
|                                          | (3) Class   | Bacteria Bacillota Negativicutes                                                                                          | 3.73 |    | 0.037   | 4.03    |
|                                          | (4) Order   | Bacteria Bacillota Negativicutes Acidaminococcales                                                                        | 3.71 |    | 0.037   | 4.01    |
|                                          | (5) Family  | Bacteria Bacillota Negativicutes Acidaminococcales Acidaminococcaceae                                                     | 3.71 |    | 0.037   | 4.01    |
|                                          | (6) Genus   | Bacteria Bacillota Negativicutes Acidaminococcales Acidaminococcaceae Phascolarctobacterium                               | 3.71 |    | 0.037   | 4.01    |
|                                          | (7) Species | Bacteria Bacillota Negativicutes Acidaminococcales Acidaminococcaceae Phascolarctobacterium Phascolarctobacterium faecium | 3.71 |    | 0.037   | 4.01    |
|                                          | (2) Phylum  | Bacteria Bacteroidota                                                                                                     | 4.60 |    | 0.046   | 4.94    |
|                                          | (3) Class   | Bacteria Bacteroidota Bacteroidia                                                                                         | 4.64 |    | 0.037   | 4.94    |
|                                          | (4) Order   | Bacteria Bacteroidota Bacteroidia Bacteroidales                                                                           | 4.64 |    | 0.037   | 4.94    |
|                                          | (5) Family  | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae                                                            | 4.63 |    | 0.037   | 4.93    |
|                                          | (6) Genus   | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides                                                | 4.31 |    | 0.037   | 4.61    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides Bacteroides fragilis                           | 3.38 |    | 0.037   | 3.68    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides Bacteroides stercoris                          | 3.99 |    | 0.037   | 4.29    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides Bacteroides thetaiotaomicron                   | 3.05 |    | 0.037   | 3.35    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides Bacteroides uniformis                          | 3.43 |    | 0.037   | 3.73    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Bacteroides Bacteroides xylanisolvens                      | 3.51 |    | 0.037   | 3.81    |
|                                          | (6) Genus   | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Phocaicola                                                 | 4.35 |    | 0.037   | 4.65    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Phocaicola Phocaicola dorei                                | 3.16 |    | 0.037   | 3.46    |
|                                          | (7) Species | Bacteria Bacteroidota Bacteroidia Bacteroidales Bacteroidaceae Phocaicola Phocaicola vulgatus                             | 4.32 |    | 0.037   | 4.62    |
|                                          | (5) Family  | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae                                                 | 4.35 |    | 0.037   | 4.65    |
|                                          | (6) Genus   | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae Parasutterella                                  | 4.35 |    | 0.037   | 4.65    |
|                                          | (7) Species | Bacteria Pseudomonadota Betaproteobacteria Burkholderiales Sutterellaceae Parasutterella Parasutterella excrementihominis | 4.35 |    | 0.037   | 4.65    |